AGG Standard Data format

What is an S-N curve?

An S-N curve (Stress–Number of cycles curve) describes the relationship between the applied cyclic stress and the number of cycles required to cause fatigue failure of a material.

Each fatigue test provides a pair of values:

- the maximum cyclic stress σ_{max}
- the number of cycles to failure N
- For the Sendeckyj model, the residual strength σ_r can also be used when no failure occurs.

These experimental data are used to construct the S-N diagram and fit fatigue models. In the S-N Curve module, three modelling approaches are available:

- **Lin-Log**
- **Log-Log**
- **Sendeckyj**

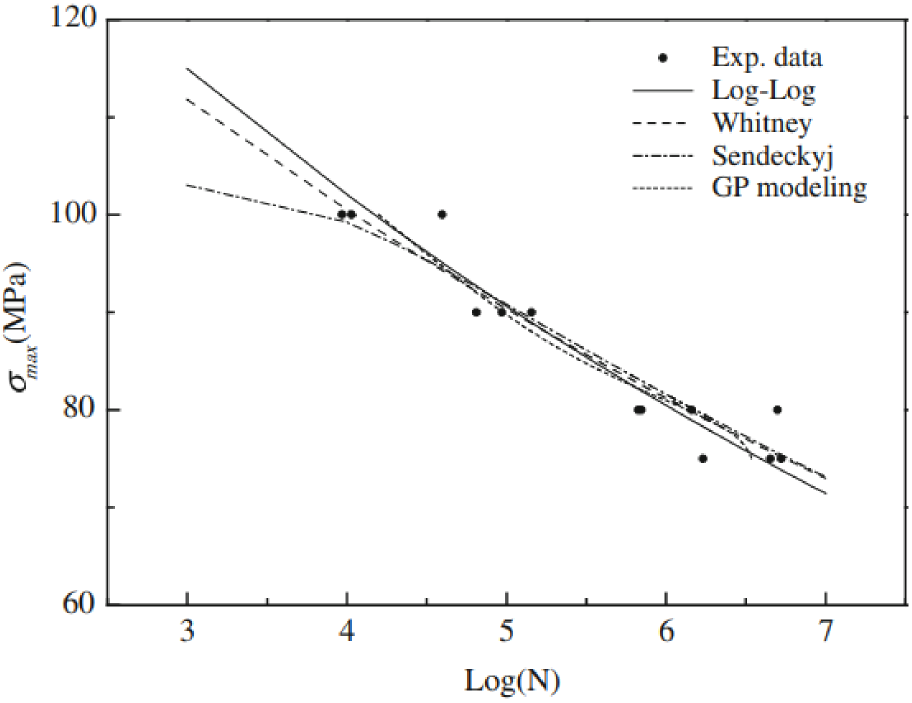


Figure 1 : Schematic representation of an S-N curve showing experimental fatigue data and fitted fatigue models (Vassilopoulos & Keller, 2017).

What is an AGG file?

An AGG file contains the experimental fatigue data used as input for the S-N Curve module. Each row corresponds to one fatigue test and provides the information needed to relate the applied cyclic stress to the number of cycles to failure.

The S-N analysis is performed together with additional information:

- an ASTM reference table (ASTM.csv) containing statistical coefficients used for regression calculations, available from ASTM reference (pp. 20–23)
- a reliability level parameter $P(N)$, defined by RELIABILITY_LEVEL, with values in the interval [0,1] and a default value of 0.5

The reliability level defines the survival probability associated with the predicted S-N curve.

File naming convention

File: AGG_{Researcher's lastname}_{Date}_{Test type}.csv

- YYYY-MM = starting month of the experiment
- Example: AGG_Dupont_2024-03_SN.csv

File format requirements

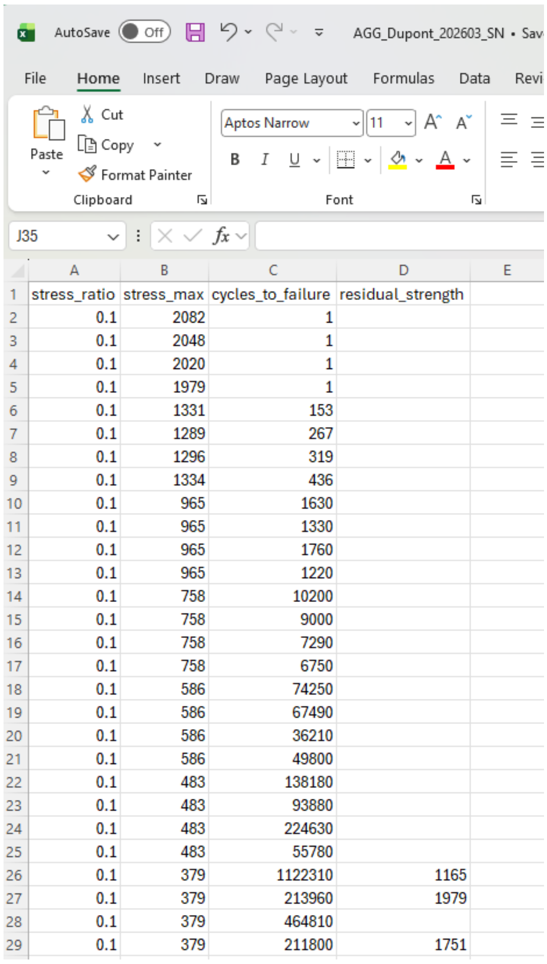
- Encoding: UTF-8
- Separator: comma (,)
- File type: .csv
- Header required
- Units: MPa

Required column (column names must be exact)

Variable name	Description	Symbol	Unit	Data type	Mandatory
stress_ratio	Stress ratio wijer	R	[-]	double	y
stress_max	Max stress	sigma_max	[MPa]	double	y
cycles_to_failure	Number of cycles to failure	N	[-]	int	y
residual_strength	Residual strength for run-out specimens. Leave empty for failure. Use -1 if the specimen is a run-out but no residual strength was measured. -1 is supported by Whitney, but not by Sendeckyj.	sigma_r	[MPa]	double	y for Sendeckyj and Whitney

Example of valid LDS file

Test values



stress_ratio	stress_max	cycles_to_failure	residual_strength
0.1	2082	1	
0.1	2048	1	
0.1	2020	1	
0.1	1979	1	
0.1	1331	153	
0.1	1289	267	
0.1	1296	319	
0.1	1334	436	
0.1	965	1630	
0.1	965	1330	
0.1	965	1760	
0.1	965	1220	
0.1	758	10200	
0.1	758	9000	
0.1	758	7290	
0.1	758	6750	
0.1	586	74250	
0.1	586	67490	
0.1	586	36210	
0.1	586	49800	
0.1	483	138180	
0.1	483	93880	
0.1	483	224630	
0.1	483	55780	
0.1	379	1122310	1165
0.1	379	213960	1979
0.1	379	464810	
0.1	379	211800	1751

AGG file

For the AGG file, Microsoft Excel works well for editing and saving the data.
However, other tools such as LibreOffice Calc, Google Sheets, or any text editor (e.g., Notepad++, VS Code) can also be used, provided the file is saved in CSV format (UTF-8 encoding).